

Locating the Next Tranche:

A denominator, a clock, a scale, a price and a fee schedule for the item panel

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Abstract

The paper is moving to a real-options frame, and that changes what the data has to do. The outcome of interest becomes the *filing* decision, which needs a denominator — the parcels that could have produced a case and did not — and the developer’s clock starts at filing, not at the first hearing. The item table is a numerator measured from the wrong instant. This memo finds, verifies, caches and measures the five categories of data that fix that, and records the negatives so nobody runs these searches again.

Five results.

First, the parcel risk set exists and the panel is shorter than the corpus. DataSF’s parcel layer holds 236,556 parcels, active and retired, on a unique `blklot` key, and 12,195 of the 13,620 items that carry a parcel key — 89.5% — join to it. That is below the 95% the brief asked for, and the shortfall is not formatting: 1,232 items name a block that exists and a lot that never did. The value side is the assessor’s secured roll (`wv5m-vpq2`, the proposed identifier, verified): 3,934,467 parcel-years over 222,251 parcels — but roll years 2007–2025, not 1998–2026. **9 years of the corpus have no assessed value at all**, and that is the binding constraint on the risk set, not a scrape artefact. Zoning is better than expected: 9 vintages, 1998–2008, are published *parcel-keyed* and join directly; 11 more are polygons.

Second, the case number joins the Planning Department’s records at 92.7%, and that buys the pre-hearing wait. Raw string equality reaches only 74.7%; stripping the case-number suffix reaches 92.7%. For the 8,334 cases that join, **the median wait from filing to first hearing is 142 days**, the 90th percentile 758 and the 99th 2346. The delay memo reports a 245-day median from hearing to permit issuance, with a 90th percentile of 768, and has nothing before the hearing. The wait it could not see has a smaller median and essentially the same 90th percentile — so, relative to its own centre, it is the more dispersed of the two, which is exactly the quantity a real-options threshold responds to.

Third — the most consequential finding — the building-permits bridge works, and it overturns the conditions memo. That memo (§5.1) concludes that the permit route reaches discretionary review and the conditions route reaches conditional use, and that “they never cover the same item.” Following a conditional use through its planning record to the permit numbers the department itself attaches, **2,336 of 5,074 conditional-use items — 46.0% — reach a permit that is present in DBI**, against the 0.3% that print one in the minutes. The non-overlap is not a property of the world; it was a property of the one route we had. Across the whole docket 54.0% of items now reach a real building permit by some route.

Fourth, project scale is reachable but thin where it matters. A unit count is attached to 91.9% of conditional-use items through the Projects table — and is *non-zero* on 27.1%. The purpose-built pipeline datasets are worse: the Development Pipeline reaches 4.5% and is a current snapshot, and Housing Production reaches 17.6% by parcel.

Fifth, the price side is stronger than expected, and it joins on the parcel key. The CoreLogic/Cotality extract is on disk under the data root: **415,854 arms-length priced San**

Francisco sales, 1968–2024, over 174,102 parcels. Cotality prints San Francisco’s APN as block and lot, so it is not a block-group price — it is `blklot`, the same key as `§2`, on 100.0% of transactions. **56.7% of items carrying a parcel key sit on a parcel that has traded**, and 22.9% on one that traded within 3 years of the hearing itself. Zillow and the assessor’s values are cached behind it as fallbacks. On fees, 8 annual Citywide Development Impact Fee Registers, 2019–2026, parsed to 1,341 dated rate observations keyed by Planning Code section — the Transportation Sustainability Fee for residential projects over 99 units runs \$10.29 to \$13.39 per gross square foot over those 8 years, and the inclusionary fee \$199.50 to \$249.66 per square foot.

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1 What this memo is for, and what “found” means in it

A developer sinks entitlement cost before the hearing outcome resolves. Dispersion in delay and in conditions raises the rent threshold at which filing is worth doing at all, so the observable consequence of a demanding Commission is not worse projects but *fewer and higher-value* ones. Two things follow for the data, and everything below serves one of them.

1. The outcome is the **filing decision**, so the analysis needs a population at risk. The item table is a numerator: it holds the requests that were made. The denominator is every parcel in the city that could have produced a request and did not.
2. The developer’s clock starts at **filing**. Every delay measure in the pipeline today starts at the first hearing, which is well downstream of the decision being modelled.

Found, usable, and the difference. This memo reports a source as *usable* only when the join key is present *and* the join rate has been measured against the item table. A file that downloaded is not a result. Where a dataset’s documentation promises a field and the rows are empty, the empty rate is reported and not the documentation — Table 8 is that table, and §4 contains the sharpest instance of it.

Two identifier facts that will bite anyone reusing this. Both were found by the probe stage and neither could have been assumed.

- **DataSF’s Socrata domain is now `data.sf.gov`.** The old `data.sfgov.org` still answers the *resource* endpoint — which is why `analyze_permits.py` keeps working — but the *catalogue* API indexes only the new domain. A catalogue search against the old host returns zero hits and looks exactly like “the dataset does not exist.” Every dataset in this memo was resolved against the live catalogue on 2026-09-08 and the identifier reported in Table 1 is the one actually used.
- **wv5m-vpq2 is the right assessor identifier** — the brief’s proposed ID, verified — **but the roll starts at 2007**. That is not a gap in a scrape; no earlier vintage is published. It is reported as a negative in Table 13 because the temptation to go looking again will be strong.
- **A Dropbox subtree that has never been materialised is invisible to `ls` and `find`.** The macOS file provider holds files online-only: the directory entry carries the full size while the file occupies no blocks, and an un-materialised directory does not list at all. The first pass of this memo concluded on that basis that the CoreLogic extract was gone. It was not — it was dehydrated, and §5 is what it actually contains. `acquire_external_data.py sync` now reports nominal bytes against bytes on disk, which is the only way to tell the two cases apart.

Table 1 is the inventory: resolved identifier, host, row count, distinct count on the key we join on, whether that key is actually unique, and the date range. Key uniqueness is tested rather than assumed, because the last time it was assumed — DBI’s `permit_number` — 145,795 rows turned out to repeat one, once per lot the permit touched. Three of the sources here are also not unique on their intended key and are marked as such.

Table 1: Every identifier resolved against the live catalogue on 2026-09-08. ‘Key unique’ is tested on the cached rows, not assumed: DBI’s `permit_number` was not unique and 145,795 rows repeated one. A dash means the column is not a key we join on.

Dataset	Identifier	Rows	Distinct key	Key unique	Date range
<i>Fees</i>					
Neighborhood-Specific Impact Fee Areas	ntc3-dd64	33	33	yes	—
<i>Filing dates</i>					
Planning Department Records — Non-Projects	y673-d69b	233,394	233,394	yes	1920-04-07–2026-09-04
Planning Department Records — Projects	qvu5-m3a2	54,515	54,515	yes	1974-05-21–2026-09-04
<i>Parcel risk set</i>					
Assessor Historical Secured Property Tax Rolls	wv5m-vpq2	3,934,467	222,251	no (3,712,216 repeats)	2007–2025
Parcels — Active and Retired	acdm-wktn	236,556	236,556	yes	1984-07-31–2026-09-02
San Francisco Land Use	c5ge-t6pj	153,765	153,754	no (11 repeats)	2026-08-17–2026-08-17
<i>Project scale</i>					
Dwelling Unit Completion Counts by Building Permit	j67f-aayr	2,541	2,420	no (121 repeats)	2002-04-01–2205-03-13 (<i>sic</i>)
Housing Production — 2005–present	xdht-4php	6,139	5,146	no (993 repeats)	2005-01-03–2026-08-28
MOHCD Affordable Housing Pipeline	aaxw-2cb8	190	131	no (59 repeats)	1995-09-19–2030-03-15
San Francisco Development Pipeline	6jgi-cpb4	1,917	1,793	no (124 repeats)	1997-06-12–2026-07-28
<i>Zoning (parcel-keyed)</i>					
Historic Zoning Districts — 1998	piqy-pimd	152,569	152,535	no (34 repeats)	—
Historic Zoning Districts — 2000	itx2-wzp5	153,292	153,247	no (45 repeats)	—
Historic Zoning Districts — 2001	8m6w-8yuk	153,669	153,593	no (76 repeats)	—
Historic Zoning Districts — 2002	3yi7-eyfd	153,663	153,582	no (81 repeats)	—
Historic Zoning Districts — 2003	6ru7-zq4y	153,716	153,518	no (198 repeats)	—
Historic Zoning Districts — 2004	s3eh-jdwb	191,062	190,896	no (166 repeats)	—
Historic Zoning Districts — 2005	b52k-gy2v	196,344	192,031	no (4,313 repeats)	—
Historic Zoning Districts — 2007	pe87-v8tx	153,822	153,751	no (71 repeats)	—
Historic Zoning Districts — 2008	tsdh-z53a	153,355	153,355	yes	—

Table 2: The parcel join, and what the residual is. The item table’s key is `assessor_block` plus each entry of `lot_number`, zero-padded to DataSF’s 4+3 `blklot` form; an item joins if any of its lots does. Percentages are of the 13,620 items that carry a parcel key at all.

Items	Count	Share
All items in the corpus	16,199	100.0%
carrying a block and at least one lot	13,620	84.1%
joined to <code>blklot</code> in the parcel layer	12,195	89.5%
... to a parcel still active	9,115	66.9%
... to a retired parcel only	3,080	22.6%
joined to <code>mapblklot</code> (the surviving map parcel)	12,038	88.4%
not joined: block exists, that lot does not	1,232	9.0%
not joined: no such block	193	1.4%

Table 3: The assessor roll as a parcel-year panel, and how far it reaches. The panel is unbalanced by construction — a parcel enters when it is created and leaves when it is merged away — and it starts at roll year 2007, which is the binding constraint on the risk set.

Quantity	Value
Rows (parcel-years)	3,934,467
Distinct parcels	222,251
Roll years	2007–2025 (19 years)
Distinct item parcel keys sought	8,294
found on the roll	5,565
median roll years observed for one of them	19 of 19
present in every roll year	4,372
Items with a parcel key joined to the roll	11,112 (81.6%)

Table 4: Zoning, by how it is published. The parcel-keyed years join to the item table directly; the polygon years need a spatial join and are inventoried, not cached. 1999 is not published in either form.

Year	Identifier	Rows	Distinct blklot	Form
1998	piqy-pimd	152,569	152,535	parcel-keyed (blklot)
1999	—	0	—	not published
2000	itx2-wzp5	153,292	153,247	parcel-keyed (blklot)
2001	8m6w-8yuk	153,669	153,593	parcel-keyed (mapblklot)
2002	3yi7-eyfd	153,663	153,582	parcel-keyed (blklot)
2003	6ru7-zq4y	153,716	153,518	parcel-keyed (blklot)
2004	s3eh-jdwb	191,062	190,896	parcel-keyed (blklot)
2005	b52k-gy2v	196,344	192,031	parcel-keyed (blklot)
2006	afkh-hfhr	23,132	—	polygon; needs a spatial join
2007	pe87-v8tx	153,822	153,751	parcel-keyed (blklot)
2008	tsdh-z53a	153,355	153,355	parcel-keyed (blklot)
2009	jud5-ja46	23,355	—	polygon; needs a spatial join
2010	x4gj-zjx7	9,897	—	polygon; needs a spatial join
2011	rt4q-mf68	9,880	—	polygon; needs a spatial join
2012	vfz8-awmy	9,889	—	polygon; needs a spatial join
2013	6jb9-g73z	9,982	—	polygon; needs a spatial join
2014	hg44-eza7	9,981	—	polygon; needs a spatial join
2015	ada2-cu6t	9,978	—	polygon; needs a spatial join
current (districts)	xzez-p3nc	1,647	—	polygon; needs a spatial join
current (height and bulk)	usp2-teig	1,195	—	polygon; needs a spatial join
current (special use)	rxfc-8aap	125	—	polygon; needs a spatial join

Table 5: Does a planning record’s identifier join to the item table’s case number? Tested three ways over all 8,987 distinct cases. ‘Stem’ strips the letter suffix (2014.0400CUA→2014.0400), which is how a suffixed case reaches its umbrella project record.

Join	Cases matched	Rate
Raw string equality	6,716	74.7%
Normalised (upper-case, whitespace stripped)	6,736	75.0%
Suffix-stripped stem	8,334	92.7%
Either	8,334	92.7%
heard 1998–2004	2,025 / 2,613	77.5%
heard 2005–2009	1,754 / 1,776	98.8%
heard 2010–2014	1,547 / 1,557	99.4%
heard 2015–2019	1,450 / 1,476	98.2%
heard 2020–2026	1,558 / 1,565	99.6%
<i>The residual, by the format the minutes print the case number in</i>		
YY.NNN	536 unmatched of 536	0.0% matched
YYYY–NNNNNN	8 unmatched of 2,648	99.7% matched
YYYY.NNNN	92 unmatched of 5,786	98.4% matched
other	17 unmatched of 17	0.0% matched

Table 6: Filing to first hearing, in days, for cases that join a planning record with an **open_date**. One observation per case. The negative tail is real and is reported rather than trimmed: a record opened after the hearing is a record created for a follow-on action, and it is a reason to use the *earliest* open date of the case, which is what is done here.

Cases	<i>N</i>	Coverage	Median	90th pct	99th pct	Negative
All cases	8,987	92.7%	142	758	2346	0.3%
conditional use	3,055	93.7%	175	608	1824	0.0%
discretionary review	2,453	92.4%	105	327	960	0.3%
variance	550	96.5%	280	870	2347	0.0%
planning code amendment	728	96.2%	64	409	1841	0.4%
heard 1998–2009	4,393	86.1%	126	580	1491	0.4%
heard 2010–2017	2,359	98.7%	169	944	3027	0.3%
heard 2018–2026	2,235	99.5%	161	894	2330	0.3%

Table 7: The `building_permits` bridge, by request type. A permit is reached if the case’s own planning record, its parent project record, or any record sharing its stem names one; ‘in DBI’ means the number, normalised to digits, is present in the cached Building Permits table. The permit-number column is the route `analyze_permits.py` already had, for comparison, and the last column is the union of the two — the share of the docket that now reaches a real building permit by some route.

Request type	Items	Reaches a permit	... present in DBI	Printed permit number	Either route
<code>discretionary_review</code>	4,117	72.4%	71.8%	85.5%	93.2%
<code>large_project_authorization</code>	308	70.5%	70.5%	0.0%	70.5%
<code>variance</code>	1,172	63.5%	63.3%	1.6%	63.9%
<code>downtown_project</code>	321	57.3%	57.3%	1.9%	57.9%
<code>conditional_use</code>	5,074	46.3%	46.0%	0.3%	46.1%
<code>other</code>	459	44.0%	44.0%	0.7%	44.2%
<code>office_allocation</code>	301	41.9%	41.9%	0.0%	41.9%
<code>ceqa_environmental</code>	652	40.6%	40.6%	0.0%	40.6%
<code>appeal_preliminary_negative_declaration</code>	537	36.5%	36.5%	0.0%	36.5%
<code>conditional_use_modification</code>	295	33.2%	33.2%	2.0%	35.3%
<code>general_plan_amendment</code>	334	32.0%	32.0%	0.0%	32.0%
<code>informational</code>	587	23.0%	23.0%	0.7%	23.2%
<code>historic</code>	177	20.9%	20.9%	0.0%	20.9%
<code>rezoning_map_amendment</code>	261	20.7%	20.7%	0.0%	20.7%
<code>planning_code_amendment</code>	1,301	7.7%	7.6%	0.0%	7.6%
All items	16,199	48.6%	48.4%	22.1%	54.0%

Table 8: What the two planning-records tables carry that the item table does not, with the share of rows on which the field is actually populated. Documentation is not evidence: these are the rows.

Field	Table	What it is	Populated	Rows
<code>open_date</code>	non-project	date the record was opened	100.0%	233,394
<code>close_date</code>	non-project	date it was closed	94.9%	233,394
<code>record_status</code>	non-project	application status	100.0%	233,394
<code>applicant</code>	non-project	applicant name	66.9%	233,394
<code>applicant_org</code>	non-project	applicant organisation	34.3%	233,394
<code>assigned_to_planner</code>	non-project	case planner	77.7%	233,394
<code>building_permits</code>	non-project	associated permit numbers	22.9%	233,394
<code>parent_id</code>	non-project	the umbrella project record	26.7%	233,394
<code>number_of_units_prop</code>	project	proposed dwelling units	100.0%	54,515
<code>number_of_units_net</code>	project	net new dwelling units	94.0%	54,515
<code>number_of_affordable_units</code>	project	affordable units	100.0%	54,515
<code>project_decision</code>	project	the department’s decision	18.3%	54,515
<code>project_decision_date</code>	project	decision date	17.3%	54,515
<code>environmental_document_type</code>	project	CEQA document type	25.1%	54,515
<code>building_permits</code>	project	associated permit numbers	45.2%	54,515
<code>inclusionary</code>	project	inclusionary flag	2.9%	54,515

Table 9: Project scale: what each source offers, and what share of the 5,074 conditional-use items in the item table it actually reaches. ‘Join key available’ is a property of the file; the last column is the measurement.

Source	Unit field	Join key	Years	CU items reached
Planning Records — Projects	number_of_units_ prop	case-number stem	1974–2026	91.9% populated, 27.1% non-zero
SF Development Pipeline	net_pipeline_ units	case_no (PRJ) + blklot	1997–2026	4.5%
Housing Produc- tion 2005–	net_units	ppts_project_id + blocklot	2005–2026	11.2% by case, 17.6% by parcel
MOHCD Afford- able Pipeline	total_project_ units	planning_case_ number	1995–2030	1.5%
Unit Completion Counts	number_of_units_ certified	building permit number	2002– 2205 (<i>sic</i>)	via the permit bridge only

Table 10: Prices and rents. The first row is the one that matters: a licensed transaction file, already on disk, that joins to the item table on the same parcel key as everything else in §2. The rows below it are the fallbacks, in descending order of quality, and are what an analysis would be left with if the licence lapsed.

Source	Status	Grain	What it gives
CoreLogic / Cotality	cached, licensed	transaction × parcel	415,854 arms-length priced San Francisco sales 1968–2024 over 174,102 parcels, joined on blklot
CoStar	institutional ac- cess question	property and lease	commercial rents; not attempted, not scraped
Zillow ZHVI	cached	26 San Francisco ZIPs, monthly	2000-01 to 2026-07, 319 months
Zillow ZORI	cached	24 San Francisco ZIPs, monthly	2015-01 to 2026-07, 139 months
Assessor as- sessed values	cached	parcel-year	land and improvement value, 2007–2025; Proposition 13 makes it a function of tenure
ACS rents and values	handled by demand_ estimation	tract	median rent and value; the demand pipeline collects it

Table 11: The Citywide Development Impact Fee Register, one per year. ‘Rate observations’ counts every dollar figure the register prints; ‘sections’ is the number of distinct Planning Code (or other code) sections those rates are keyed to. The effective date is read out of the register’s own cover line, not inferred from when the file was retrieved.

Register	Pages	Rate observations	Code sections	Effective
2019	10	172	27	January 1, 2019
2020	11	165	27	January 1, 2020
2021	11	167	28	January 11, 2021
2022	11	168	28	January 1, 2022
2023	11	168	28	January 1, 2023
2024	11	168	28	January 1, 2024
2025	12	157	28	January 1, 2025
2026	12	176	28	January 1, 2026

Table 12: Rates behind the largest obligation-imposing condition headings in the conditions memo, as each register prints them, keyed by Planning Code section and by the register’s own land-use band. Dollars per gross square foot except §415, which the register states per square foot of residential gross floor area. A dash is a band the register of that year does not print. These are the rates in force in January of the register year; the Article 4 reduction adopted in August 2026 post-dates the last register cached here.

Fee	2019	2020	2021	2022	2023	2024	2025	2026
Transportation Sustainability Fee, residential 21–99 units	\$9.11	\$9.61	\$9.95	\$10.55	\$11.18	\$11.40	\$11.63	\$11.86
Transportation Sustainability Fee, residential >99 units	\$10.29	\$10.86	\$11.24	\$11.91	\$12.62	\$12.87	\$13.13	\$13.39
Transportation Sustainability Fee, non-residential 800–99,999 gsf	\$21.23	\$22.40	\$23.18	\$24.57	\$26.04	\$26.56	\$27.09	\$27.63
Inclusionary Affordable Housing, §415 (per sq ft of residential)	\$199.50	\$210.47	\$217.84	\$230.91	\$244.76	\$249.66	\$249.66	\$249.66
Jobs-Housing Linkage Program, §413	\$28.57	\$28.13	\$29.11	\$30.86	\$32.71	\$33.36	\$34.03	\$34.71
Child Care, residential §414	\$1.85	\$1.95	\$2.02	\$2.14	\$2.27	\$2.32	\$2.37	\$2.41
Child Care, commercial §414A	\$2.15	\$2.27	\$2.35	\$2.49	\$2.64	\$2.69	\$2.74	\$2.80
Eastern Neighborhoods Infrastructure, §423	\$24.00	\$25.32	\$26.21	\$27.78	\$29.45	\$30.04	\$30.64	\$31.24
School Impact Fee, residential (State Ed. Code)	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79

Table 13: Negative results. A row here means the search was run and is not to be run again. The first two are carried over from the conditions memo and were not re-run.

What was sought	Where it was looked for	Result
Condition-of-approval text	DBI Building Permits (i98e-djp9)	87 of 1,295,048 rows mention one — conditions memo §4, not re-run
A commission-actions dataset	DataSF catalogue	does not exist — conditions memo §4, not re-run
Assessor roll before 2007	wv5m-vpq2	the roll starts at 2007; no earlier vintage is published
Parcel-keyed zoning for 1999	DataSF historic zoning series	1998 and 2000 exist; 1999 does not
Parcel-keyed zoning after 2008	DataSF historic zoning series	2009–2015 are polygon layers only
A time-varying height and bulk layer keyed on parcel	DataSF historic height and bulk series	2009–2014 published as polygons; no parcel-keyed vintage
A market price after 2024-05-16	\$MFHR_DATA_ROOT/demand/corelogic	the Cotality pull ends there, two years short of the corpus; Zillow covers the tail at ZIP level only
Impact fee registers before 2019	sfplanning.org and the Wayback Machine	the register is republished at one URL; no snapshot before 2019-09-15
A unit count for the early docket	every scale source in Table 9	a non-zero count reaches 18.3% of conditional uses heard before 2005 against 30.9% after

2 The parcel risk set

2.1 Geometry and identifiers

The layer is **Parcels — Active and Retired** (`acdm-wktn`): 236,556 rows, one per parcel that has ever existed in the city’s records, unique on `blklot`, and carrying `mapblklot` (the surviving map parcel a retired lot was folded into), an `active` flag, and four dated fields — when the parcel was added to and dropped from the recorder’s file and from the map. That last group is what makes it a risk set rather than a snapshot: a parcel can enter and leave, and the dates say when.

The item table’s key is `assessor_block` plus each entry of `lot_number`, zero-padded to DataSF’s four-plus-three `blklot` form. 13,620 items — 84.1% — carry one, covering 8,294 distinct parcels.¹

Table 2 is the join, and the residual is characterised rather than dropped:

- **89.5% join** — 12,195 of 13,620.
- 66.9% join to a parcel that is *still active*; 3,080 items join only to a parcel that has since been retired. This is the substantively interesting part of the residual, not noise: a lot that was merged away after its hearing is frequently a lot that was merged *because* of it.
- **1,232 items name a block that exists and a lot that does not.** That is the bulk of the failure and it is not a formatting problem — the block matched.
- 193 items name no block the city has ever recorded.

The brief’s acceptance criterion was a join on $\geq 95\%$ of items carrying a parcel key. **It is not met: the rate is 89.5%.** Saying so is cheaper than discovering it in a hazard model. The upper panel of Figure 1 shows that the rate is flat in time — this is not an era effect — so the shortfall is a property of how the minutes print lots, and the 1,232 block-matched failures are the population worth a hand pass.

2.2 Value: the assessor’s secured roll

Assessor Historical Secured Property Tax Rolls (`wv5m-vpq2`) is the value side: 3,934,467 parcel-years over 222,251 distinct parcels, carrying assessed land value, improvement value, fixtures, use code and definition, property class, year built, lot area, property area, number of units and stories — a parcel-year panel in the shape the hazard needs. Table 3 gives it.

The constraint is the period. **The roll runs 2007–2025 (19 years)**, and the corpus runs 1998–2026. The lower panel of Figure 1 shows the roll is flat inside its window — a median of 208,927 parcels a year, every year, with no hole — so this is a genuine start date and not the 2018 problem again. Of the 8,294 item parcel keys, 5,565 appear on the roll at some point, and 81.6% of items with a parcel key reach it. Of the item parcels that do appear, 4,372 appear in every one of the 19 years; the rest enter or leave, so the panel is unbalanced by construction and has to be treated that way.

¹The corpus memo reports 8,295 distinct parcels for the same corpus. The difference is one key: an item with a block and an *empty* lot, which the corpus memo’s rule zero-pads into the plausible-looking parcel 0168000. A blank lot is not lot 0, so it is dropped here.

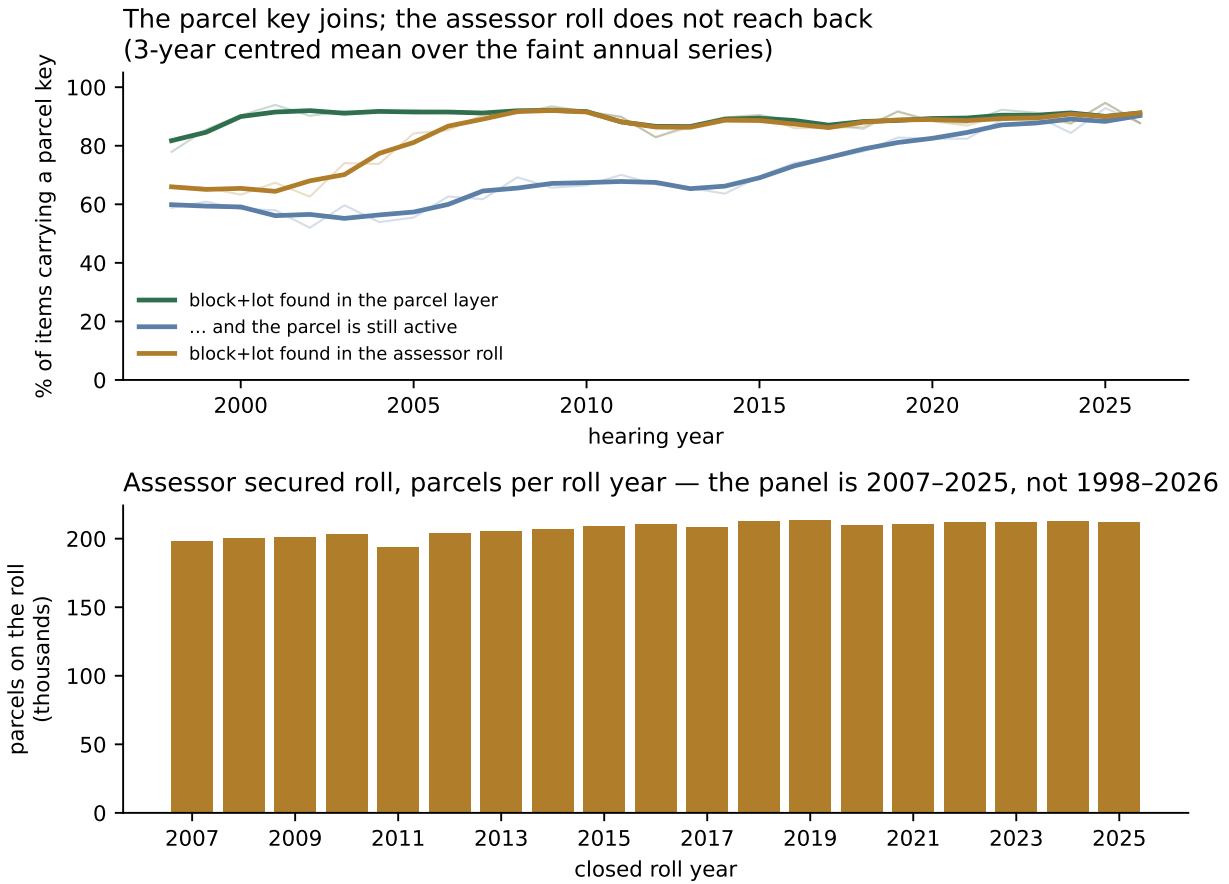


Figure 1: Top: the share of items carrying a parcel key that join the parcel layer, that join a parcel still active, and that appear on the assessor’s roll, by hearing year. Bottom: parcels on the assessor’s secured roll per roll year — the panel is 2007–2025, and the bars are flat, so nothing is missing *within* it.

What this costs. A parcel-year hazard can be estimated for 2007–2025. That is within the brief’s acceptance range (2005–2026 was asked for, 1998 was the ideal) but it excludes the first 9 years of the corpus, which is precisely the era in which the corpus memo’s substitution — denial falling to zero, conditioning tripling — begins. Anything that wants pre-period variation in assessed value will have to come from somewhere else, and §5 is the only candidate.

2.3 Zoning, and the one genuinely good surprise

The brief asked whether any *time-varying* zoning exists, expecting a current snapshot. It does, and the shape of the publication is the finding. Table 4 is the inventory.

- **9 vintages, spanning 1998–2008, are published *parcel-keyed*,** one row per `blklot` with the zoning district, the height limit, the land use and, in the earlier years, unit counts and building square footage. These join to the item table on the same key as everything else in §2 and need no spatial work at all.
- **11 more — the remaining years and the three current layers (districts, height and bulk, special use) — are polygon-only.** They are inventoried in Table 4 with row counts and identifiers and deliberately not cached: turning them into a parcel panel is a spatial join, which this memo does not do and which should not be done accidentally.
- **One year in the middle of the parcel-keyed run is not published in either form,** and Table 4 names it.

There is also a free current snapshot: `acdm-wktn` carries `zoning_code` and `zoning_district` on every parcel. So current zoning is parcel-keyed, 1998–2008 is parcel-keyed, and the years between them are the polygon-only gap in the middle. That is an odd shape but it is a far better one than a snapshot, and it means pre-period zoning variation is available for exactly the era the assessed values are not.

Height and bulk is worse. The historic height-and-bulk series is polygon-only in every year it is published; there is no parcel-keyed vintage (Table 13). Height therefore has to come either from the 1998–2008 zoning tables’ `height_lim` column or from a spatial join.

Land use (`c5ge-t6pj`) is available as a cross-check on the assessor’s use codes, with one wrinkle: its key is `mapblklot`, and 77 of its rows describe a *group* of parcels (`geography_type` = `multiple_parcel`s, with a list of `mapblklot` values in one cell) rather than a single parcel. That is few enough to handle as an exception rather than as a design constraint.

3 Filing dates and the conditional-use-to-permit bridge

3.1 Does the case number join?

The two Planning Department Records tables — `y673-d69b` (Non-Projects, 233,394 rows) and `qvu5-m3a2` (Projects, 54,515 rows) — are the department’s own application record. Both identifiers were verified and both are unique on `record_id`. Table 5 tests the join three ways against all 8,987 distinct cases in the item table.

- **Raw string equality: 74.7%.**
- **Normalised (upper-case, whitespace stripped): 75.0%.** Normalisation buys almost nothing here, which is a small vindication of the extraction — the case numbers come out clean.

- **Suffix-stripped stem: 92.7%.** Stripping the letter suffix (2014.0400CUA → 2014.0400) is what does the work, because it is how a suffixed case reaches its umbrella project record. 8,334 of 8,987 cases join.

Figure 2 gives it by hearing year, and the residual has one cause. **Coverage does not start late; it starts abruptly.** From 2002 onward the annual join rate never falls below 96%. Before that it collapses, and Table 5’s last block says why: the minutes of that era print the case number in the two-digit-year form (98.426D), and **the Planning Department’s records contain no identifier in that format at all — 0 of 536 such cases match.** Of the modern-format cases only 92 fail to join. The residual is a format, not a coverage rate, and no amount of fuzzy matching fixes it: the records simply do not go back that far under any key.

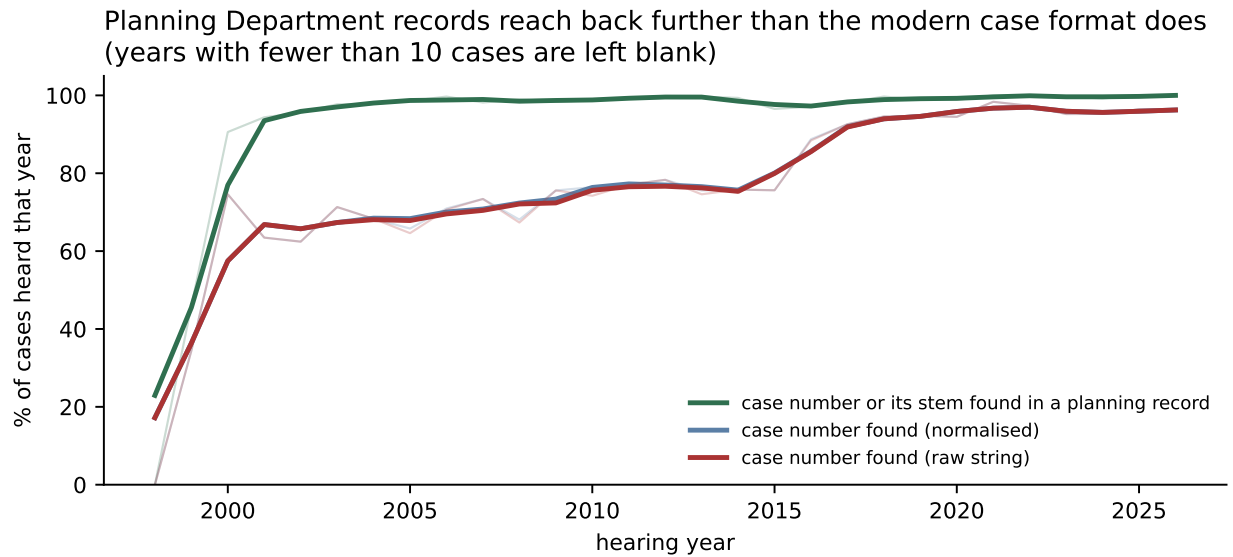


Figure 2: Share of cases heard each year whose case number is found in a Planning Department record, three ways. The early collapse is the two-digit-year case-number format, which the department’s records do not use.

3.2 What open_date buys: the wait before the hearing

open_date is populated on 100.0% of non-project records (Table 8), so every joined case has one. Collapsing both sides to one row per case — earliest open date, earliest hearing — gives the elapsed time a developer actually experiences before the Commission first looks at the project. Table 6 and Figure 3 report it.

- **Median 142 days.** 90th percentile 758; 99th 2346. Coverage 92.7% of all 8,987 cases.
- By request type the ordering is the one the theory wants: **variance 280 days, conditional use 175, discretionary review 105.** A discretionary review is filed against someone else’s pending permit and is heard quickly; a conditional use is the developer’s own application and waits.
- 0.3% of observations are negative — a record opened after the hearing. Those are follow-on records, and they are the reason the *earliest* open date of the case is the one used. They are reported rather than trimmed.

This is the memo’s answer to the delay memo’s blind spot. That memo measures hearing-to-issuance at a 245-day median with a 768-day 90th percentile — both recomputed here from the artefact that memo wrote, and both reproduced exactly. The pre-hearing wait is a median of 142 days with a 90th percentile of 758: a shorter typical wait and an almost identical tail, and therefore a markedly larger ratio of tail to centre. A real-options model in which dispersion is what raises the filing threshold has been ignoring the more dispersed of the two waits, and starting its clock after the more dispersed one had already run.

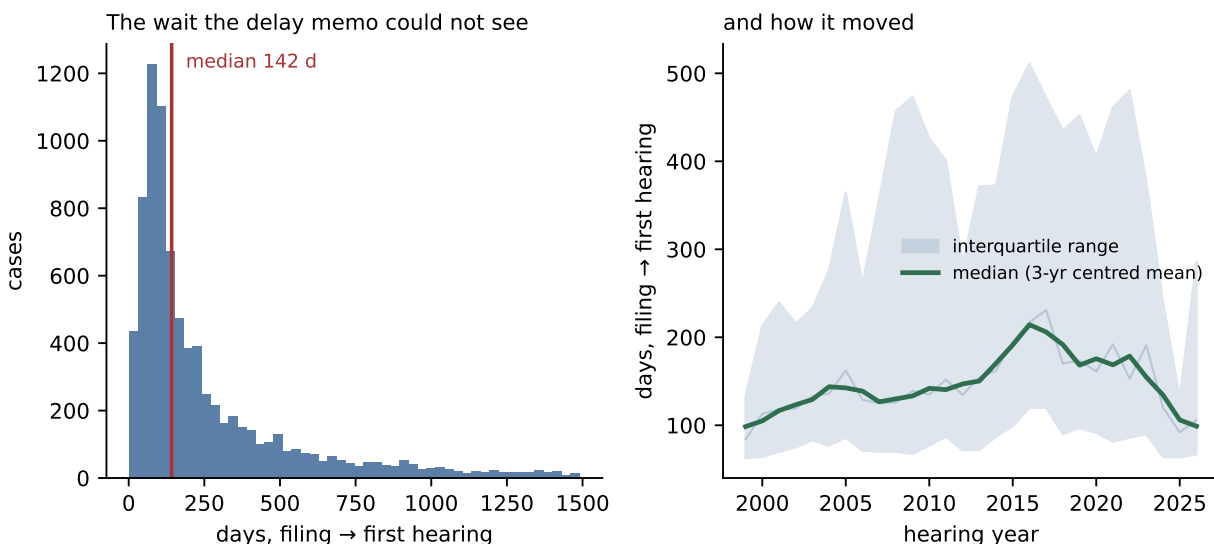


Figure 3: Left: the distribution of days from filing to first hearing, truncated at 1,500 days for display; the median line is computed on the untruncated data. Right: the median by hearing year with the interquartile range behind it, years with fewer than 10 cases omitted.

3.3 The bridge, and a correction to the conditions memo

This was the highest-value question in the brief and the answer is yes.

`building_permits` is a field on both records tables, populated on 22.9% of non-project rows and 45.2% of project rows (Table 8). For each item, the permit numbers were read from the case’s own record, from its parent PRJ record where `parent_id` names one, and from any record sharing its stem; each was normalised to digits exactly as `analyze_permits.py` normalises a printed one — no zero-padding, because DBI does not pad the serial — and matched against the cached Building Permits table. Table 7 is the result.

- **Conditional use: 46.0% of 5,074 items reach a permit present in DBI**, against 0.3% that print a permit number in the minutes. The route the permit memo had was blind to this part of the docket almost completely; the bridge is not.
- Discretionary review reaches 71.8% through the bridge — *lower* than the 85.5% that print a number — so the two routes are complements, not substitutes. Taking either, 93.2% of discretionary reviews reach a permit.
- **Across the whole docket, 54.0% of items reach a building permit by one route or the other**, against 48.4% by the bridge alone and 22.1% by the printed number alone.

- Large project authorization (70.5%), variance (63.3%) and downtown project (57.3%) were all effectively unreachable before and are now reachable. Legislative items (planning code amendment, 7.6%) remain unreachable, correctly — they are not about a building.

Correction. The conditions memo, §5.1, concludes:

“The permit route reaches discretionary review (83.8% of DR items match a DBI permit) and essentially nothing else; the conditions route reaches conditional use and essentially nothing else. Between them they cover most of the docket, but they never cover the same item, so an analysis that wants both construction outcomes and condition content on one project cannot have them from these two sources.”

That conclusion no longer holds. 2,336 conditional-use items — the exact population the packet route reaches — now also reach a building permit in DBI. The non-overlap was a property of the route, not of the docket: the minutes do not print a permit number on a conditional use, but the Planning Department records one anyway. Construction outcomes and condition content *can* be had on the same project, for 46.0% of conditional uses. **The conditions memo has been amended accordingly:** §5.1 now carries a dated correction paragraph quoting the sentence it withdraws, and its §8 item 5 (“the exact complement of the permit route”) has been narrowed to a claim about what the minutes print rather than about what is knowable.

3.4 What else the records carry

Table 8 inventories the fields the item table does not have, with the share of rows on which each is actually populated. `close_date` (94.9%), `record_status` (100.0%), `applicant` (66.9%), `assigned_to_planner` (77.7%) and `applicant_org` (34.3%) are all substantially populated and all are new information. Two are not what their names suggest: `project_decision` is populated on 18.3% of project rows, so the department’s own decision field is not a substitute for the extraction, and `inclusionary` is a flag on 2.9%.

4 Project scale

Table 9 is the answer, and it contains the memo’s clearest instance of the difference between a field existing and a field carrying information.

The Projects table is the spine. Its `number_of_units_prop` reaches 91.9% of conditional-use items through the case-number stem, over a date range that starts long before any purpose-built pipeline dataset. But the field is never null and is *zero* on most rows: **it is non-zero on only 27.1% of conditional-use items.** Reporting the first number alone would be reporting the documentation. The right statement is that a unit count is *attached* to the great majority of conditional uses and is *informative* on well under a third of them — which is consistent with the docket, most of which is not residential construction, but which caps what a revenue measure built on it can do.

The purpose-built datasets are weaker than the brief hoped:

- **SF Development Pipeline** (6jgi-cpb4) does carry a planning case number — `case_no` — which is what the brief flagged as a potential P0 find. It is the PRJ umbrella number, so it reaches the item table only through the stem, and it is a **current snapshot**: 1,917 rows, the projects in the pipeline now. It reaches 4.5% of conditional-use items. It is a good source for scale on live projects and not a panel.

- **Housing Production 2005–present** (xdht-4php) carries `ppts_project_id` in the old YYYY.NNNN form, which joins to the item table’s stem, and `blocklot`. It reaches 17.6% of conditional-use items by parcel. It is completions, so it observes only projects that were built.
- **MOHCD Affordable Housing Pipeline** (aaxw-2cb8) carries `planning_case_number` explicitly and is 190 rows. It reaches 1.5% of conditional-use items. It is the right source for the affordability channel and it is small.
- **Dwelling Unit Completion Counts** (j67f-aayr) is keyed on the building permit number, so it is reachable only through §3.3’s bridge — which is now a reason it is reachable at all.

And none of it reaches the early docket well. Every purpose-built scale dataset begins in 2005 or later, and the consequence is measurable rather than rhetorical: a non-zero unit count reaches 18.3% of conditional uses heard before 2005 against 30.9% of those heard after.

5 Prices and rents

Table 10 is the fallback chain. The top of it is the finding.

CoreLogic / Cotality: the pull happened, and the extract is here. The brief asked whether the pull went through before Stanford access lapsed. It did, and the files are under the data root: `demand/` holds 76 files and 4.92 GB, including both the raw Cotality extracts and the cleaned parquet `demand_estimation/corelogic.py` built from them.

An earlier pass of this memo reported the whole `demand/` subtree as absent. It was wrong, and the reason is worth recording because it will recur. Dropbox’s macOS file provider keeps files *online-only*: the directory entry reports the full size while the file occupies zero blocks, and a directory that has never been materialised does not appear in a listing at all. `find` and `ls` both said nothing was there. `acquire_external_data.py sync` is now the check — it reports nominal bytes against bytes on disk and materialises what is dehydrated — and running it turned up 4.92 GB across 76 files, of which several gigabytes were placeholders. **The lesson is the one the repo already knows in another form: when the machine and the record disagree, check the source before believing the machine.**

What it is, and that it joins at the parcel. Table 10 and Figure 4 report it. The extract holds 447,883 San Francisco transactions, of which 415,854 are arms-length with a price and 320,323 are residential, over 174,102 distinct parcels.

The join is the `find`. Cotality prints San Francisco’s APN as 0836 003 — block and lot, in the same four-plus-three form DataSF uses — so stripping the separator yields `blklot` directly. **100.0% of San Francisco transactions carry a usable parcel key**, and the price therefore joins to the item table on exactly the key the risk set of §2 is built on. This is a parcel-level price, not a market-level index, and it is the difference between having $\Pi(s)$ and having a proxy for it.

What that buys, measured rather than asserted:

- **3,882 of the 8,294 distinct parcels the item table touches (46.8%) have at least one arms-length sale.**
- **56.7% of items with a parcel key** sit on a parcel that has traded at some point — 7,727 items.

- **22.9% sit on a parcel that traded within 3 years of the hearing** (3,125 items). That is the number that matters for a filing threshold: it is the share of decisions for which a *contemporaneous* market price on the actual parcel is observable, rather than a neighbourhood average.

The right panel of Figure 4 shows the first of those rising from roughly half to three quarters over the period, as more of the housing stock turns over at least once inside the window the extract covers, while the contemporaneous-trade share sits flat near a quarter. The flatness is the honest read: turnover, not data coverage, is what caps it.

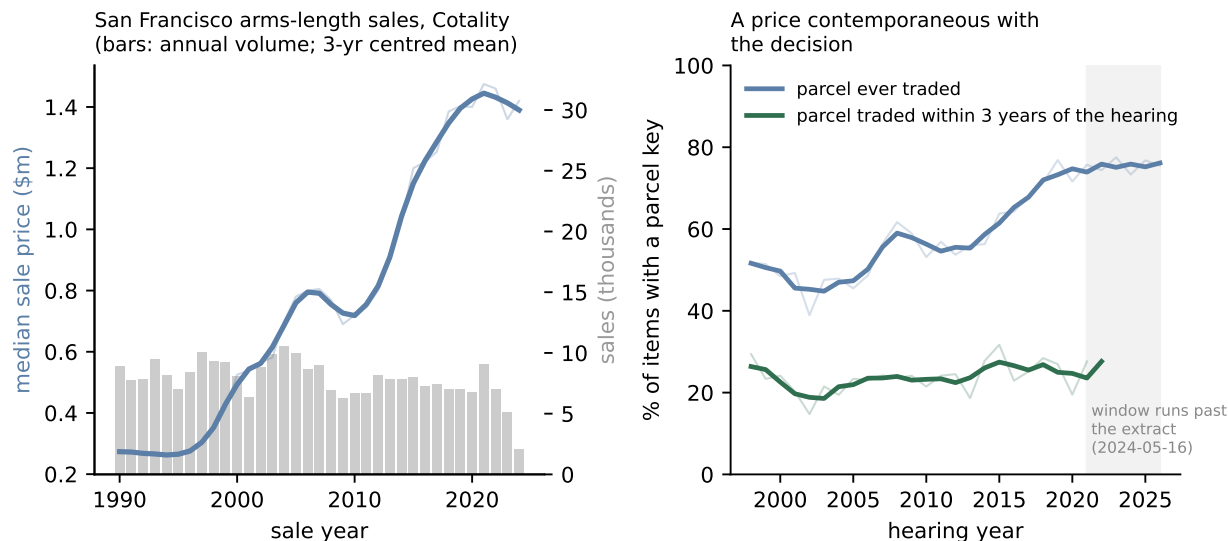


Figure 4: Left: median arms-length sale price in San Francisco with annual volume behind it, from the year the extract becomes dense. Right: the share of items carrying a parcel key whose parcel has ever traded, and whose parcel traded within 3 years of the hearing. The shaded band is the hearing years whose forward window runs past the end of the extract; the contemporaneous series is not drawn there, because a fall would be an artefact of where the file stops.

Where it stops. The extract ends 2024-05-16, two years short of the corpus, and the series is only dense from 1968 (a handful of deeds carry placeholder dates back to 1900). The last three hearing years therefore cannot have a 3-year forward window and are drawn as a gap in Figure 4 rather than as a decline — the same rule the corpus memo applies to 2018.

CoStar is an access question and is reported as one. Nothing was attempted and nothing was scraped.

What is cached instead. Zillow’s research files, filtered to San Francisco County and stored with the national row counts recorded so the filter is auditable: **ZHVI on 26 ZIP codes, 2000-01 to 2026-07**, and **ZORI on 24 ZIP codes, 2015-01 to 2026-07**. These are market-level indices; the item table has no ZIP, so using them means going parcel → address → ZIP, or falling back to the supervisor district and analysis neighbourhood the parcel layer already carries. They are a price *environment*, not a project price.

The fallback. Assessed land and improvement value from §2, parcel-year, 2007–2025, universal within that window. It is a weak price: Proposition 13 makes assessed value a function of how long the current owner has held the parcel rather than of what the parcel is worth. It is worth having with that caveat stated in every table it appears in, and it is not a substitute for a transaction price.

6 Fee and exaction schedules

The conditions memo finds that most staff-imposed conditions are obligation-imposing rather than project-changing, and that the largest recurring headings — Transportation Sustainability Fee, inclusionary, impact fees — have posted dollar schedules. This section converts those headings into dollars.

The document. The **San Francisco Citywide Development Impact Fee Register** is a multi-page landscape schedule, republished each January, giving for every fee: the city area subject to it, the ordinance reference, the uses it applies to, the rate, the threshold, the controlling entity, the unit, and the effective date. It is exactly the object the brief asked for.

The problem, and the route around it. The register is republished at *one URL*, so the department’s site only ever holds the current year. 8 dated copies were assembled — 2019 through 2026 — from the Wayback Machine’s snapshots of that URL and from the year-stamped URLs the department left behind. **No snapshot exists before 2019-09-15**, under that URL or any other tried, so the fee panel starts at 2019 (Table 13).

The parse. Table 11 reports what came out of each register: 1,341 dated rate observations in total, keyed to the Planning Code (or State Education Code) section of the register row they sit in, and to the land-use or size band the register prints on the same line. The section is the key because it is the only column that survives PDF text extraction intact; the fee *name* wraps across several lines and the columns interleave. Effective dates are read from the register’s own cover line, not inferred from when the file was retrieved.

Table 12 is the result: **a rate keyed by year, by fee and, where the register labels it, by land use**. The Transportation Sustainability Fee for residential projects over 99 units runs \$10.29 per gross square foot in 2019 to \$13.39 in 2026; the inclusionary fee under §415 runs \$199.50 to \$249.66 per square foot of residential gross floor area. A named condition heading in a Commission motion can now be priced.

What is not here. The register’s *district* dimension is a free-text column (“Eastern Neighborhoods Mixed Use Districts: RED, RED-MX, SPD, MUG...”) and is not parsed into a key. Neighborhood-Specific Impact Fee Areas (ntc3-dd64, 33 rows) is cached and is the geographic side of that, but joining a fee area to a parcel is a spatial join and is not done here. The inclusionary *requirement percentages* under §415 — as distinct from the in-lieu fee — and their amendment history are legislative and are not in the register; that remains open.

7 Established, and not established

Established.

1. A parcel-year risk set can be built. `acdm-wktn` is unique on `blklot` and 89.5% of items carrying a parcel key join to it; the residual is 1,232 block-matched lot failures and 193 unmatched blocks, not a formatting problem.
2. The value side of that risk set is 2007–2025 and no earlier. 3,934,467 parcel-years, flat across the window, 81.6% of keyed items reached.
3. Zoning is time-varying and parcel-keyed for 9 vintages spanning 1998–2008, and polygon-only for 11 more.
4. The case number joins the Planning Department’s records on 92.7% of 8,987 cases, via the suffix-stripped stem, and never below 96% in any year from 2002 on. The earlier residual is entirely the two-digit-year case-number format.
5. Filing to first hearing is measurable on 92.7% of cases: median 142 days, 90th percentile 758, 99th 2346.
6. The `building_permits` bridge reaches 46.0% of conditional-use items through to a permit present in DBI, and 54.0% of the docket reaches a permit by some route. The conditions memo’s non-overlap claim (§5.1) is overturned.
7. A unit count is attached to 91.9% of conditional-use items and is non-zero on 27.1%.
8. The CoreLogic extract is present, and it is a *parcel-level* price: 415,854 arms-length San Francisco sales over 174,102 parcels, keyed on `blklot` on 100.0% of transactions. 56.7% of items with a parcel key sit on a parcel that has traded, 22.9% on one that traded within 3 years of the hearing.
9. An annual dollar fee schedule exists for 2019–2026, keyed by Planning Code section and land-use band, 1,341 rate observations.

Not established.

1. Whether the 1,232 block-matched, lot-unmatched items are extraction errors or lots the recorder never carried. The block matched, so both are live; a hand pass on a sample would settle it and has not been done.
2. Whether a case’s *earliest open_date* is the filing date a developer would recognise. A case can have several records opened under one stem — environmental, preliminary project assessment, the entitlement itself — and the earliest is the earliest *administrative* act, which may pre-date the application the hearing is about. The 2346-day 99th percentile is the visible symptom.
3. Whether the permit a conditional use reaches through the bridge is the permit the hearing was *about*. The bridge returns every permit the department associates with the project; no adjudication of which one is the entitlement’s permit has been done, and the permit memo’s parcel-window-type rule is the obvious tool for it.
4. Whether `number_of_units_prop` means the same thing across the Projects table’s full span. It is zero on most rows and the zero is not distinguishable from an unfilled field.
5. Anything about fee levels before 2019. The register panel starts there and the earlier schedules are not on the site or in the archive under any URL tried.

6. Whether a sale near the hearing is a sale *caused* by it, or how to treat the parcels that never trade. 22.9% of keyed items have a contemporaneous price; the complement is not missing at random — a parcel that has not changed hands in decades is a different kind of parcel — and nothing here corrects for that.
7. Anything causal. Every number in this memo is a coverage rate or a descriptive distribution.

8 Limitations

Coverage rates resting on a small denominator. The MOHCD pipeline is 190 rows and reaches 1.5% of conditional-use items — that is a handful of items, and a rate computed on it should not be reported to one decimal place anywhere but the table it comes from. The Development Pipeline’s 4.5% rests on 1,917 rows. The request-type breakdown in Table 7 suppresses types with fewer than 100 items, and Table 6 suppresses groups with fewer than 10 cases with a lag.

Current snapshots that are not panels. The Development Pipeline is the projects in the pipeline *now*; a project entitled a decade ago and long since built is not in it. The current zoning, height-and-bulk and special-use layers are snapshots. `acdm-wktn`’s `zoning_code` is current zoning attached to every parcel including retired ones, which is a snapshot wearing a panel’s clothes and must not be used as the zoning *at* the time of an early hearing — that is what the 1998–2008 vintages are for. San Francisco Land Use is one `data_as_of` date.

Periods that do not line up. The corpus is 1998–2026. Assessed values are 2007–2025. Parcel-keyed zoning is 1998–2008. Fee rates are 2019–2026. Transaction prices are 1968–2024 and stop at 2024-05-16. **No single window has all of them**, and a specification wanting both an assessed value and the zoning in force is confined to the overlap of the second and third unless the polygon zoning years are joined spatially.

Access questions left open. CoStar (BU Questrom holdings, unchecked by design). The §415 inclusionary requirement percentages and their amendment history. Fee registers before 2019. CoreLogic is *not* on this list any more — the extract is on disk — but the licence is Stanford’s and lapsed, so the file cannot be refreshed past 2024-05-16 without new access. Treat what is cached as frozen.

Source quirks worth carrying. Dwelling Unit Completion Counts contains a `date_issued` in 2205, marked (*sic*) in Tables 1 and 9; it is a typo in the source, not the end of the file’s coverage. The MOHCD pipeline’s dates run to 2030 because they are estimated construction completions, which is correct. 14 of the 19 cached sources are not unique on their intended key and are marked so in Table 1. The Planning Department’s non-project table has an `open_date` in 1920, which is real in the sense that the row says so and has not been audited.

One measurement not attempted. The brief asks for a sample of non-joining parcel keys with the reason. Table 2 gives the reasons as a partition — retired-only, block-matched-lot-unmatched, no-such-block — which is the aggregate version of that, and the sample of individual keys is left for the hand pass in “Not established” item 1.

9 Reproducing this

code/commission_minutes_processing/acquire_external_data.py runs in three stages, each cached so no network stage repeats by accident:

```
python acquire_external_data.py sync          # materialise online-only files
python acquire_external_data.py probe         # resolve every identifier, live
python acquire_external_data.py fetch        # download and cache each source
python acquire_external_data.py fetch --refresh # re-download everything
python acquire_external_data.py fetch --only assessor # or one source
python acquire_external_data.py report       # joins, figures, tables, README
```

`sync` materialises every online-only file under `$MFHR_DATA_ROOT/demand/` and prints nominal bytes against bytes on disk. Run it first: it is the difference between a subtree that is missing and one that Dropbox has simply not downloaded, and this memo got that wrong once.

`probe` writes `$MFHR_DATA_ROOT/external/_acquisition/probe.json` — the resolved identifier, row count, column list and update time for every dataset, the catalogue searches that produced them, and an HTTP check on every non-Socrata file. `fetch` writes one directory per source under `$MFHR_DATA_ROOT/external/` (`parcels`, `assessor`, `zoning`, `planning_records`, `pipeline`, `prices`, `fees`) and skips anything already on disk. `fetch --only corelogic` is the exception that downloads nothing: it slices the licensed Cotality extract already under `demand/` down to San Francisco, attaches the parcel key, and caches the result, so `report` does not re-read two gigabytes. `report` regenerates the three figures, `tables/acquisition_tables.tex`, `tables/acquisition_macros.tex`, and `external/README.md`, which records per source the resolved identifier, retrieval date and row count so a later run can tell whether the cache is stale.

The item table is the same extraction the other memos read (`$MFHR_DATA_ROOT/extraction/corpus_v2.g3/clean/*.jsonl`), and the permit normalisation and the printed-permit parser are imported from `analyze_permits.py` rather than re-implemented, so the two memos cannot report different numbers for the same question.

No number in this memo is hand-placed, and here that is literal rather than conventional. Every measurement in the prose is a `\newcommand` defined in `tables/acquisition_macros.tex`, written by `report`; the tables are `tables/acquisition_tables.tex`, written by the same function; both are `\input` at the top of the `.tex`. Changing a figure in this document means changing the script.

The only numerals typed into the `.tex` are: section and item cross-references; identifiers and dataset names (2014.0400CUA, 98.426D, Housing Production 2005–present); the register’s own printed land-use bands (“over 99 units”); named prior facts (Proposition 13, the 2018 hole in the meeting corpus); the brief’s own $\geq 95\%$ acceptance criterion; and three figures quoted from sibling memos, each attributed where it appears — the corpus memo’s 8,295 distinct parcels, the permit memo’s 145,795 duplicate `permit_number` rows, and the 83.8% inside the block quotation from the conditions memo. The delay memo’s hearing-to-issuance median is *not* in that list: `report` recomputes it from `permit_matches.csv` and it appears here as `\acqDelayIssueMedian`, which is one way of checking that this memo and that one still agree.